

Anexo 15 — Interfaz cliente (HTML/CSS)

Archivo: Scripts/web_remote/static/index.html

Para que sirvió:

Página de control con estética Star Wars: vista del robot en vivo, pad para mover el objetivo, y botones de maniobras, cámara y apariencia. Pensada para celular en horizontal.

```
1 | <!doctype html>
2 | <html lang="es">
3 | <head>
4 | <meta charset="utf-8">
5 | <meta name="viewport" content="width=device-width, initial-scale=1, maximum-scale=1, user-scalable=no">
6 | <title>DUM Pit Droid – Holocrón de Control</title>
7 | <style>
8 |   :root {
9 |     --bg: #05060a;
10 |    --fg: #d6e4f0;
11 |    --gold: #ffe81f;      /* amarillo logo Star Wars */
12 |    --saber-blue: #4da6ff; /* sable Jedi */
13 |    --saber-red: #ff3b3b;  /* sable Sith / Imperio */
14 |    --mute: #5e6b7a;
15 |    --panel: #0c1018;
16 |    --panel-edge: #1d2a3a;
17 |  }
18 |  * { box-sizing: border-box; }
19 |  html, body {
20 |    margin: 0; padding: 0;
21 |    background: var(--bg);
22 |    color: var(--fg);
23 |    font-family: "Bahnschrift", "Eurostile", "Arial Narrow", system-ui, sans-serif;
24 |    -webkit-tap-highlight-color: transparent;
25 |    height: 100vh; overflow: hidden;
26 |    letter-spacing: 0.3px;
27 |  }
28 |  /* Campo de estrellas sutil de fondo */
29 |  body::before {
30 |    content: ""; position: fixed; inset: 0; z-index: 0; pointer-events: none;
31 |    background-image:
32 |      radial-gradient(1px 1px at 20% 30%, #ffffff55, transparent),
33 |      radial-gradient(1px 1px at 70% 60%, #ffffff44, transparent),
34 |      radial-gradient(1px 1px at 40% 80%, #ffffff33, transparent),
35 |      radial-gradient(1px 1px at 85% 15%, #ffffff44, transparent),
36 |      radial-gradient(1px 1px at 55% 45%, #ffffff22, transparent),
37 |      radial-gradient(2px 2px at 10% 70%, #ffe81f33, transparent);
38 |    background-repeat: repeat;
39 |  }
40 |
41 |  #app {
42 |    position: relative; z-index: 1;
43 |    display: flex; flex-direction: row;
44 |    height: 100vh; width: 100vw;
45 |  }
46 |
47 |  /* --- Panel IZQUIERDO: stream del robot --- */
48 |  #left-panel {
49 |    flex: 0 0 50%;
50 |    background: #000;
51 |    display: flex; flex-direction: column;
52 |    border-right: 1px solid var(--gold);
53 |    box-shadow: 2px 0 18px #ffe81f22;
54 |  }
55 |  #left-header {
56 |    padding: 10px 14px; background: var(--panel);
57 |    display: flex; justify-content: space-between; align-items: center;
58 |    border-bottom: 1px solid var(--panel-edge);
59 |  }
60 |  #left-header h1 {
61 |    margin: 0; font-size: 17px; font-weight: 700;
62 |    color: var(--gold); text-transform: uppercase; letter-spacing: 3px;
63 |    text-shadow: 0 0 8px #ffe81f88;
64 |  }
65 |  #status {
66 |    font-size: 11px; padding: 3px 10px; border-radius: 2px;
67 |    background: #000; border: 1px solid var(--panel-edge);
68 |    text-transform: uppercase; letter-spacing: 1px;
69 |  }
70 |  #status.ok { color: var(--saber-blue); border-color: var(--saber-blue); text-shadow: 0 0 6px #4da6ff88; }
71 |  #status.err { color: var(--saber-red); border-color: var(--saber-red); text-shadow: 0 0 6px #ff3b3b88; }
72 |  #stream-wrap { flex: 1; overflow: hidden; position: relative; background: #000; }
73 |  #robot-view { width: 100%; height: 100%; object-fit: cover; display: block; }
74 |  #state-banner {
```

```

75 | padding: 8px 12px; background: var(--panel);
76 | text-align: center; font-size: 12px;
77 | border-top: 1px solid var(--panel-edge);
78 | text-transform: uppercase; letter-spacing: 1.5px;
79 | }
80 | #state-banner .label { color: var(--mute); margin-right: 8px; }
81 | #state-banner .val { font-weight: 700; color: var(--mute); }
82 | #state-banner .val.active { color: var(--gold); text-shadow: 0 0 6px #ffe81f88; }
83 |
84 | /* --- Panel DERECHO: acciones, scrollable --- */
85 | #right-panel {
86 |     flex: 1; overflow-y: auto;
87 |     padding: 14px;
88 |     display: flex; flex-direction: column; gap: 14px;
89 | }
90 | #right-panel::-webkit-scrollbar { width: 5px; }
91 | #right-panel::-webkit-scrollbar-thumb { background: var(--gold); border-radius: 3px; opacity: 0.5; }
92 |
93 | .section-title {
94 |     font-size: 11px; color: var(--gold);
95 |     text-transform: uppercase; letter-spacing: 2px;
96 |     margin: 6px 2px 0 2px; opacity: 0.85;
97 |     border-left: 2px solid var(--gold); padding-left: 8px;
98 | }
99 |
100 | /* pad MAS CHICO + gutter scrollable a los costados */
101 | #pad-wrap {
102 |     background: var(--panel); border: 1px solid var(--panel-edge);
103 |     border-radius: 4px; padding: 14px 0; /* sin padding lateral: el ancho lo da el pad */
104 |     display: flex; justify-content: center;
105 | }
106 | #pad {
107 |     width: 80%; /* mas chico que antes (era 100%) */
108 |     aspect-ratio: 1 / 1;
109 |     background: #03040a; border: 1px solid var(--panel-edge);
110 |     border-radius: 4px; touch-action: none; display: block;
111 |     box-shadow: inset 0 0 24px #4da6ff22;
112 | }
113 | /* el #pad-wrap (los costados del pad) tiene touch-action normal -> ahi se scrollea */
114 |
115 | .slider-row {
116 |     display: flex; align-items: center; gap: 10px;
117 |     background: var(--panel); border: 1px solid var(--panel-edge);
118 |     padding: 10px 12px; border-radius: 4px;
119 | }
120 | .slider-row label { min-width: 60px; color: var(--mute); font-size: 11px;
121 |     text-transform: uppercase; letter-spacing: 1px; }
122 | .slider-row input { flex: 1; accent-color: var(--gold); }
123 | .slider-row .val { min-width: 78px; text-align: right;
124 |     font-variant-numeric: tabular-nums; font-size: 12px; color: var(--saber-blue); }
125 |
126 | #info-row { display: flex; gap: 8px; flex-wrap: wrap; }
127 | .info-cell {
128 |     flex: 1; min-width: 80px;
129 |     background: var(--panel); border: 1px solid var(--panel-edge);
130 |     padding: 8px 10px; border-radius: 4px;
131 | }
132 | .info-cell .label { color: var(--mute); font-size: 10px;
133 |     text-transform: uppercase; letter-spacing: 1px; }
134 | .info-cell .val { font-size: 18px; font-weight: 700; margin-top: 2px;
135 |     font-variant-numeric: tabular-nums; color: var(--saber-blue); }
136 | #focus-cell .val.YES { color: var(--gold); text-shadow: 0 0 6px #ffe81f88; }
137 | #focus-cell .val.no { color: var(--mute); }
138 |
139 | /* Botones estilo "panel de mando" */
140 | .action-btn {
141 |     padding: 16px; font-size: 15px; font-weight: 700;
142 |     border: 1px solid; border-radius: 4px; cursor: pointer;
143 |     touch-action: manipulation; background: #000;
144 |     text-transform: uppercase; letter-spacing: 2px;
145 |     transition: box-shadow .15s, background .15s;
146 | }
147 | #grab-btn { color: var(--saber-red); border-color: var(--saber-red); }
148 | #grab-btn:not(:disabled):active { background: #1a0606; box-shadow: 0 0 16px #ff3b3b99; }
149 | #wave-btn { color: var(--saber-blue); border-color: var(--saber-blue); }
150 | #wave-btn:not(:disabled):active { background: #06121a; box-shadow: 0 0 16px #4da6ff99; }
151 | .action-btn:disabled { opacity: 0.35; cursor: default; }
152 |
153 | /* Maniobras: acciones a la izquierda, D-pad de camara a la derecha */
154 | #maniobras-row { display: flex; gap: 8px; }
155 | #maniobras-actions { flex: 2; display: flex; flex-direction: column; gap: 8px; }
156 | #maniobras-cam { flex: 1.1; display: flex; align-items: center; justify-content: center; }
157 | .dpad {
158 |     display: grid;
159 |     grid-template-columns: repeat(3, 1fr);

```

```

160 |     grid-template-rows: repeat(3, 1fr);
161 |     gap: 4px; width: 100%; height: 100%; min-height: 112px;
162 | }
163 | .dpad-up { grid-column: 2; grid-row: 1; }
164 | .dpad-left { grid-column: 1; grid-row: 2; }
165 | .dpad-right { grid-column: 3; grid-row: 2; }
166 | .dpad-down { grid-column: 2; grid-row: 3; }
167 | .dpad-eye {
168 |     grid-column: 2; grid-row: 2;
169 |     display: flex; align-items: center; justify-content: center;
170 |     font-size: 20px; color: var(--gold); opacity: 0.85;
171 |     text-shadow: 0 0 8px #ffe81f88;
172 | }
173 | .cam-btn {
174 |     font-size: 16px; font-weight: 700;
175 |     background: #000; color: var(--gold); border: 1px solid var(--gold);
176 |     border-radius: 4px; cursor: pointer; touch-action: manipulation;
177 |     display: flex; align-items: center; justify-content: center;
178 | }
179 | .cam-btn:active { box-shadow: 0 0 12px #ffe81f88; background: #14110a; }
180 |
181 | /* Botones de apariencia (skin) */
182 | #skin-row { display: flex; gap: 8px; }
183 | .skin-btn {
184 |     flex: 1; padding: 12px; font-size: 13px; font-weight: 700;
185 |     background: #000; color: var(--mute); border: 1px solid var(--panel-edge);
186 |     border-radius: 4px; cursor: pointer; touch-action: manipulation;
187 |     text-transform: uppercase; letter-spacing: 1.5px;
188 |     transition: all .15s;
189 | }
190 | .skin-btn.skin-active {
191 |     color: var(--gold); border-color: var(--gold);
192 |     box-shadow: 0 0 12px #ffe81f55; text-shadow: 0 0 6px #ffe81f88;
193 | }
194 |
195 | @media (orientation: portrait) {
196 |     #app { flex-direction: column; }
197 |     #left-panel { flex: 0 0 40vh; border-right: none; border-bottom: 1px solid var(--gold); }
198 |     #right-panel { padding: 12px; }
199 | }
200 | </style>
201 | </head>
202 | <body>
203 |
204 | <div id="app">
205 |
206 |     <!-- IZQUIERDA: stream fijo -->
207 |     <div id="left-panel">
208 |         <div id="left-header">
209 |             <h1>DUM &middot; Pit Droid</h1>
210 |             <span id="status" class="err">sin enlace</span>
211 |         </div>
212 |         <div id="stream-wrap">
213 |             
214 |         </div>
215 |         <div id="state-banner">
216 |             <span class="label">estado</span>
217 |             <span class="val" id="state-val">IDLE</span>
218 |         </div>
219 |     </div>
220 |
221 |     <!-- DERECHA: acciones scrollables -->
222 |     <div id="right-panel">
223 |
224 |         <div class="section-title">Maniobras</div>
225 |         <div id="maniobras-row">
226 |             <div id="maniobras-actions">
227 |                 <button id="grab-btn" class="action-btn">Atrapar esfera</button>
228 |                 <button id="wave-btn" class="action-btn">Saludar</button>
229 |             </div>
230 |             <div id="maniobras-cam">
231 |                 <div class="dpad">
232 |                     <button id="cam-up" class="cam-btn dpad-up">▲</button>
233 |                     <button id="cam-left" class="cam-btn dpad-left">◀</button>
234 |                     <span class="dpad-eye">&#128065;</span>
235 |                     <button id="cam-right" class="cam-btn dpad-right">▶</button>
236 |                     <button id="cam-down" class="cam-btn dpad-down">▼</button>
237 |                 </div>
238 |             </div>
239 |         </div>
240 |
241 |         <div class="section-title">Apariencia</div>
242 |         <div id="skin-row">
243 |             <button id="skin-realista" class="skin-btn skin-active">Realista</button>
244 |             <button id="skin-imperio" class="skin-btn">Imperio</button>

```

```

245 |     </div>
246 |
247 |     <div class="section-title">Objetivo del sensor (esfera roja)</div>
248 |     <div id="pad-wrap">
249 |         <canvas id="pad" width="600" height="600"></canvas>
250 |     </div>
251 |
252 |     <div class="slider-row">
253 |         <label>Distancia</label>
254 |         <input id="dist-slider" type="range" min="-65" max="95" value="60">
255 |         <span class="val" id="dist-val">0.60 m frente</span>
256 |     </div>
257 |
258 |     <div class="section-title">Telemetría</div>
259 |     <div id="info-row">
260 |         <div class="info-cell">
261 |             <div class="label">theta</div>
262 |             <div class="val" id="theta-val">— °</div>
263 |         </div>
264 |         <div id="focus-cell" class="info-cell">
265 |             <div class="label">foco</div>
266 |             <div class="val no" id="focus-val">—</div>
267 |         </div>
268 |         <div class="info-cell">
269 |             <div class="label">fps</div>
270 |             <div class="val" id="fps-val">—</div>
271 |         </div>
272 |     </div>
273 |
274 | </div>
275 |
276 | </div>
277 |
278 | <script src="/static/control.js"></script>
279 | </body>
280 | </html>
281 |

```